

GNN Modelisations for Orbit Wars

Six graph representations for fleet decision-making

2026-06-07

Overview

1. **M1 — Edge prediction:** attacks as directed edges on planet graph
2. **M2 — Hetero attack nodes:** tripartite planet $\rightarrow$ attack $\rightarrow$ planet
3. **M3 — Attack-only graph:** attack nodes + same-src / same-dst edges
4. **M4 — Step nodes:** ETA lifted into a dedicated node type
5. **M5 — Temporal unrolling:** $N \times 10$ planet copies across time
6. **M6 — Hybrid bipartite:** base + temporal snapshot + attack nodes



planet



attack

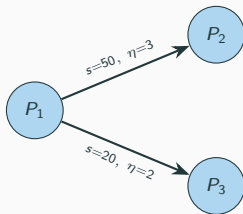


step



master

M1 — Edge Prediction on Planet Graph



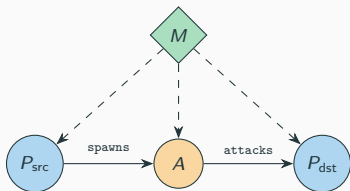
Type	Features
Planet node (22-dim)	$x/100, y/100, \text{nature}^\dagger, \text{prod}/5, \text{owne}$ $\left[\frac{\log s_t}{\log 1024} \right]_{t=0}^{10}$
Attack edge (3-dim)	$\frac{\log s_{\min}}{\log 1024}, \frac{\log s_{\max}}{\log 1024}, \eta/9$

† fix/moving/comet (3 one-hot)

$^\dagger^\dagger$ neutral + 4 players (5 one-hot)

Task: multi-label edge classification

M2 — Heterogeneous Attack Nodes



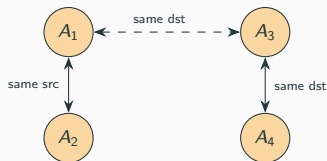
Type		Features
Planet	(22-dim)	same as M1
Attack node	(3-dim)	$\frac{\log s_{\min}}{\log 1024}, \frac{\log s_{\max}}{\log 1024}, \eta/9$
Master	(6-dim)	$\text{step}/500, \frac{\omega - 0.025}{0.025}, \text{ship-prop} \times 4$

Matches 97-library.py

build_hetero_data_97.

Task: node classification on attack nodes

M3 — Attack-Only Graph

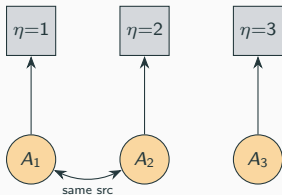


Type	Features
Attack node (3-dim)	$\frac{\log s_{\min}}{\log 1024}, \frac{\log s_{\max}}{\log 1024}, \eta/9$
Same-src edge	attacks from the same source planet
Same-dst edge	attacks targeting the same planet

No planet nodes — pure attack graph.

Task: node classification on attack nodes

M4 — Step Nodes (ETA as Structure)

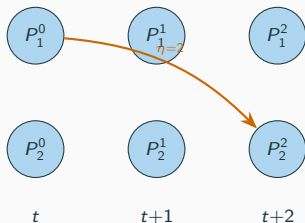


Type	Features
Attack node (2-dim)	$\frac{\log s_{\min}}{\log 1024}, \frac{\log s_{\max}}{\log 1024}$
Step node (0-dim)	no features; identity = ETA
Attack $\rightarrow$ Step	connects attack to its ETA bucket
Same-src/dst	same as M3

ETA is structural, not a feature.

Task: node classification on attack nodes

M5 — Temporal Unrolling



Type	Features
Planet ^t node (12-dim)	$x^t/100, y^t/100, \text{nature}^\dagger, \text{prod}/5,$ $\text{owner}^{\dagger\dagger}, \frac{\log s^t}{\log 1024}$
Attack edge (1-dim)	$\frac{\log s_{\min}}{\log 1024}$

[†] fix/moving/comet

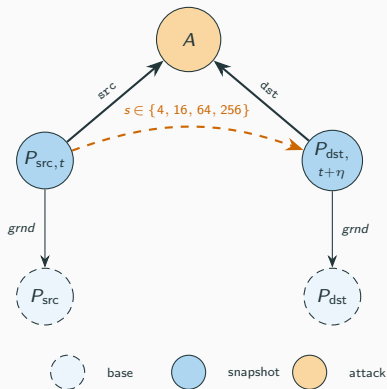
^{††} 5-class one-hot

$$P_{\text{src}}^{t=i} \xrightarrow{\eta} P_{\text{dst}}^{t=i+\eta}$$

$N \times 10$ planet nodes total.

Task: multi-label edge classification

M6 — Hybrid Bipartite Action-Selector



Type	Features
Base P_i	nature [†] (one-hot), prod/5 (4-dim)
Snapshot $P_{i,t}$	$x/100, y/100, \frac{\log s}{\log 1024},$ (9-dim) owner ^{††} (one-hot), $t/20$
Attack A	$\frac{\log s_{fleet}}{\log 1024}$ (1-dim)

[†] fix/moving/comet

^{††} 5-class one-hot

Edges:

- $P_{i,t} \rightarrow P_i$ state-to-base (grounding)
- $P_{src,t} \rightarrow A, P_{dst,t+\eta} \rightarrow A$ attack-action
- $P_{i,t_i} \dashrightarrow P_{j,t_j}$ temporal ($s \in \{4, 16, 64, 256\}$)

Task: node classification on attack nodes

Applicable to any modelisation:

- **Master node** — global context, connected to all planet/attack nodes
 - step/500, $(\omega - 0.025)/0.025$, ship-proportion $\times 4$ players
- **Comet-aware features** — comet flag in nature one-hot; optionally add path index
- **Lookahead labels** — label attacks by forward-simulated Δ_{ships} instead of heuristic selection
- **Fleet nodes** — in-flight fleets as explicit nodes:
 $x, y, \text{angle}, s, \text{ETA}_{\text{each planet}}$